

TOWARDS REVERSIBLE MODEL MERGING FOR LOW-RANK WEIGHTS

Anonymous authors

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ABSTRACT

Model merging seeks to consolidate multiple fine-tuned models into a single set of weights that performs well across all source tasks. Although prior work has established that merging can approximate the per-task performance of individually fine-tuned models, it has largely neglected the increasingly common regime in which models are stored in low-rank form—whether through low-rank adaptation (LoRA) or post-training singular value decomposition (SVD). We show that applying conventional merging methods to such low-rank weights induces severe degradation in the merged model, a failure we trace to task interference amplified by limited representational capacity and to the broken coupling between the low-rank factors. This diagnosis motivates a fundamentally different formulation: rather than collapsing all adapters into one set of weights, we construct a compact basis—equivalent in cost to retaining only a small number of models—from which each original task-specific model can be recovered by linear combination. Merging thereby becomes the construction of a reconstruction-capable model space rather than the production of a single merged model, and this reformulation makes it possible to “revert” to any individual model on demand, in recognition of the fact that no merged model can consistently outperform one specialized for its task. Building on this principle, we introduce Reversible Model Merging (RMM), an efficient, data-free, and flexible method that admits a closed-form solution for jointly selecting the optimal basis of model weights and the task-specific coefficients that govern their linear combination. Extensive experiments spanning diverse datasets, backbone architectures, and compression regimes demonstrate that RMM consistently and substantially outperforms existing merging approaches, preserving the performance of low-rank compressed models by a wide margin.

1 INTRODUCTION

Parameter-efficient fine-tuning methods, foremost among them Low-Rank Adapters (LoRA) (Hu et al., 2021), have become a cornerstone of adapting large language models to diverse downstream tasks without incurring the full cost of retraining. In parallel, post-hoc compression techniques—notably post-training singular value decomposition (SVD) truncation—are now widely adopted to curb storage and inference costs, adding a complementary axis of efficiency (Ryu et al., 2023; Ping et al., 2024; Wang et al., 2025b;a). Alongside these developments, model merging has emerged as a compelling paradigm for consolidating the knowledge of many task-specific models into a single unified set of weights. A rapidly growing body of work, spanning simple parameter interpolation through optimization-based schemes, has shown that merged models can perform competitively across source tasks without requiring access to the original training data (Matena & Raffel, 2022; Yang et al., 2024; Yadav et al., 2023; Wei et al., 2025).

Despite this progress, a fundamental limitation has gone largely unaddressed: even the most sophisticated merging strategies consistently fall short of the performance of the original individually fine-tuned models. This gap widens markedly when the models being merged have already been compressed into low-rank form, whether through LoRA fine-tuning or post-training SVD truncation. The reason is structural. By enabling a single model to serve many tasks, merging necessarily collapses distinct task-specific representations into one shared parameter space; the resulting interference erodes the specialized adaptations each model acquired during fine-tuning (Gargiulo et al.,

2025). This tension between consolidation and fidelity, we argue, is not incidental but intrinsic to the single-model formulation of merging.

Although low-rank adapters and SVD-truncated models are individually compact, their proliferation introduces a new scalability bottleneck at the collection level. In realistic multi-task, federated, and continual-learning deployments, practitioners routinely accumulate dozens or even hundreds of task-specific low-rank adapters, each demanding separate storage and coordinated management. Retaining all of them erodes precisely the storage savings that low-rank parameterization was designed to deliver and renders deployment increasingly cumbersome. This state of affairs establishes a concrete need for merging strategies that operate directly on low-rank representations—not merely to consolidate storage and reduce redundancy, but to enable the efficient sharing, deployment, and transfer of compressed models across tasks.

To ground this need in evidence, we first characterize how existing merging techniques behave when applied to low-rank compressed models. Our analysis centers on deltas—the weight differences between each fine-tuned model and the shared pre-trained initialization—which conventional merging strategies attempt to combine into a single set of parameters under an implicit assumption of linear compatibility. We show empirically that merging low-rank deltas in this manner precipitates catastrophic performance degradation. The failure is attributable to two coupled factors: the limited expressive capacity inherent to low-rank representations, and the misalignment of task-specific subspaces that, having been optimized independently, interfere destructively when superimposed. These findings indicate that the difficulty is not a tuning artifact but a systematic incompatibility between the single-model merging objective and compressed representations.

This observation prompts us to reconsider the objective of merging itself: *What if merging need not terminate in a single final model? What if the problem were relaxed to permit retaining more than one component? What if we preserved the ability to reconstruct each original model on demand?* Pursuing these questions, we propose to replace a single collapsed set of weights with a compact set of weights that functions as a shared basis, one from which every original model can be recovered by linear combination. This reframing opens an underexplored design space: instead of compressing all information into one model, we retain complementary representational directions across a small set of basis models, thereby enabling post-merging “reversion” to the original adapters. In doing so, our approach overturns the prevailing assumption that merging must yield a single model and redefines merging as the construction of a reconstruction-capable model space.

Building on this reconstruction-driven perspective, we develop a theoretical framework that characterizes the optimal basis for minimizing reconstruction error, and we instantiate it as our method, Reversible Model Merging (RMM). RMM recasts merging as a basis-selection problem whose objective is to identify a compact set of task vectors that supports accurate reconstruction of all original models. Within this framework, RMM admits a closed-form solution that simultaneously determines an optimal set of basis task vectors and the task-specific coefficients used to recover each individual adaptation by linear combination. Our contributions are as follows. First, we identify and empirically characterize the catastrophic failure of conventional merging on low-rank compressed models, tracing it to amplified task interference and the broken coupling between low-rank factors. Second, we reformulate merging as the construction of a reconstruction-capable model space and derive, via Theorem 1, a closed-form, data-free basis-selection solution. Third, across a broad range of datasets, backbone architectures, and compression regimes, we show that RMM consistently surpasses conventional merging strategies by a substantial margin while preserving the efficiency benefits of low-rank parameterization, and that a single tunable hyperparameter—the number of retained components—affords a controllable trade-off between storage and reconstruction fidelity that scales favorably to large model collections.

2 RELATED WORK

Model merging has established itself as a powerful strategy for consolidating knowledge from multiple fine-tuned models. Existing methods fall into two broad families: data-free parameter-space manipulations, and optimization-based approaches that leverage data or auxiliary objectives. We review representative techniques from both lines of work below, with particular attention to how each treats the interference that arises when task-specific updates are superimposed.

Task Arithmetic: The foundational abstraction underlying modern merging is task arithmetic (Ilharco et al., 2023), which defines a delta as the difference between the parameters of a fine-tuned model and its pre-trained initialization. Its central insight is that such deltas behave as modular, composable building blocks: they can be added to a base model to impart new abilities, subtracted to suppress undesired behaviors, or scaled and combined to blend multiple capabilities. This linear view of weight-space composition provides the conceptual foundation on which subsequent interference-aware methods build.

DARE: Although simple delta addition can be effective, it degrades when the merged models carry conflicting or overlapping parameter updates. To mitigate this, Yu et al. (2024) introduced DARE (Drop and Rescale), which relies on aggressive randomized pruning. DARE discards the vast majority of parameter changes—typically 90–99%—on the premise that most updates are inessential and may actively interfere with knowledge contributed by other models; the surviving updates are then rescaled to preserve the overall magnitude of the modification. The net effect is to retain the core task-specific components of each delta while suppressing the interference that would otherwise accompany dense superposition.

TIES: Where DARE resolves conflicts stochastically, TIES-merging supplies a deterministic three-phase procedure—Trim, Elect Sign, and Merge—that systematically resolves interference across models (Yadav et al., 2023). It first trims each delta to its largest-magnitude entries, retaining the most influential fine-tuning updates; then, at each parameter position, it elects a majority sign across models and discards contributions that oppose it; finally, it averages the sign-consistent parameters into the merged representation. This consensus-driven pipeline curtails destructive interference and yields a more stable merged model. Sharing the same interference-reduction motivation, EMR-Merging (Elect, Mask, and Rescale) elects a single unified model and attaches task-specific binary masks and rescalers, derived from each individual model, that steer the unified model toward each task at inference time (Huang et al., 2024).

Other Merging Methods: A complementary family moves beyond purely parameter-space manipulation and instead relies on explicit optimization or access to a small dataset. Model soups (Wortsman et al., 2022) refine the task-arithmetic merging weights by minimizing validation loss, steering the interpolation toward better generalization, while Fisher-weighted averaging (Matena & Raffel, 2022) injects second-order information into the merge, up-weighting parameters estimated to be more critical for task performance. Others cast merging directly as an optimization problem: RegMean (Jin et al., 2023) and AdaMerging (Yang et al., 2024) adaptively search for task-dependent merging coefficients by solving small-scale objectives—the former on the discrepancy between merged- and individual-model predictions, the latter on the entropy of predictions over unlabeled data. More recently, Wei et al. (2025) formulate merging as minimizing the loss difference between the merged and individual fine-tuned models and derive a tractable surrogate through a first-order Taylor expansion. These optimization-based methods can attain stronger task-specific performance, but they characteristically require labeled or unlabeled data, additional optimization, or both—prerequisites that restrict their applicability precisely in the data-free, retraining-free regime that motivates our work.

3 PRELIMINARIES

We formalize the problem of consolidating a collection of fine-tuned models into a single unified model through merging. Let $\{\theta_1, \dots, \theta_n\}$ denote n models fine-tuned on n distinct tasks. A merging algorithm $\mathcal{M}$ maps these models to a single set of weights $\theta_{\mathcal{M}} = \mathcal{M}(\{\theta_i\}_{i=1}^n)$. The objective is to identify a merging procedure $\mathcal{M}$ that maximizes the average validation performance across all tasks:

$$\max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\theta_{\mathcal{M}}, \mathcal{D}_i), \quad (1)$$

where $\mathcal{D}_i$ denotes a held-out validation set for task i and Perf denotes any suitable performance metric (e.g., accuracy or F1 score). Following standard practice, we assume that all fine-tuned models share a common pretrained initialization θ_{pre} . For any layer l , the update (or delta) of model i relative to this base is $\Delta_i^l = \theta_i^l - \theta_{\text{pre}}^l$, where $\Delta_i^l \in \mathbb{R}^{m \times d}$ is the weight difference for a layer of dimensions $m \times d$.

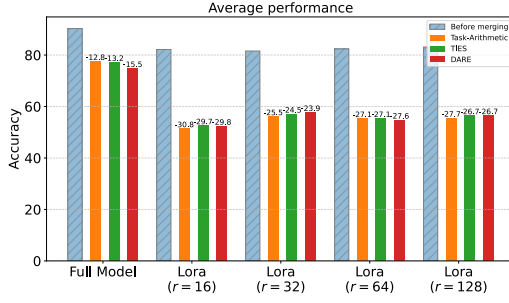


Figure 1: Average task performance of individual models (before merging) versus merged models across merging methods, for fully fine-tuned models and for LoRA-compressed models at ranks $r \in \{16, 32, 64, 128\}$. The widening gap between the individual-model reference and every merged variant under LoRA compression exposes the catastrophic degradation that motivates our approach.

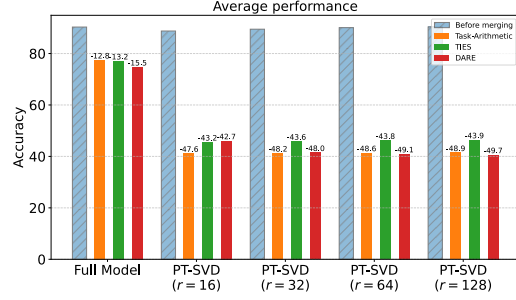


Figure 2: Average task performance of individual models (before merging) versus merged models across merging methods, for fully fine-tuned models and for PT-SVD-compressed models at ranks $r \in \{16, 32, 64, 128\}$. As with LoRA, conventional merging collapses under low-rank compression while the individual-model reference remains high, confirming that the failure is intrinsic to the compressed regime rather than to a particular compression scheme.

3.1 LOW-RANK COMPRESSED MODELS

To reduce storage as well as training and inference costs, fine-tuned models are routinely compressed into low-rank form. We consider the two most widely adopted compression strategies:

LoRA fine-tuning: During training, each layer is augmented with trainable low-rank factors $A \in \mathbb{R}^{m \times r}$ and $B \in \mathbb{R}^{r \times d}$, with $r \ll \min(m, d)$. The pretrained weights are held frozen, and optimization is confined to A and B .

Post-training SVD truncation (PT-SVD): After standard full fine-tuning, the update Δ_i^l is decomposed by SVD and truncated to its top- r singular components, yielding the low-rank approximation $\hat{\Delta}_i^l = A_i^l B_i^l$ with $A_i^l \in \mathbb{R}^{m \times r}$ and $B_i^l \in \mathbb{R}^{r \times d}$, where $r \ll \min(m, d)$.

In both regimes the fine-tuned update for layer l of model i therefore admits the low-rank factorization $\hat{\Delta}_i^l = A_i^l B_i^l$, with $A_i^l \in \mathbb{R}^{m \times r}$, $B_i^l \in \mathbb{R}^{r \times d}$, and r the compression rank. Merging thus reduces to the problem of combining low-rank updates across tasks, i.e., constructing a merged model that integrates the compressed deltas $\hat{\Delta}_i = \{\hat{\Delta}_i^l\}_{l=1}^L$ over all tasks:

$$\hat{\theta}_{\mathcal{M}} = \theta_{\text{pre}} + \mathcal{M}(\{\hat{\Delta}_i\}_{i=1}^n), \quad \max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\hat{\theta}_{\mathcal{M}}, \mathcal{D}_i), \quad (2)$$

where L is the total number of layers. Throughout this work we focus on this setting, in which every model is compressed into low-rank form. The next section investigates, both empirically and mechanistically, how conventional merging strategies behave under this low-rank regime.

4 MOTIVATING OBSERVATION

We now examine how conventional merging strategies behave when applied to the low-rank compressed models introduced above. For each layer l , prior work has considered two natural extensions of merging (Tang et al., 2025; Zhao et al., 2025; Zheng et al., 2025; Zhang et al., 2025):

- **Combined merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{A_i^l B_i^l\}_{i=1}^n),$$

in which each low-rank update is first expanded to its full-rank matrix form and only then merged.

- **Separate merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{A_i^l\}_{i=1}^n) \mathcal{M}(\{B_i^l\}_{i=1}^n),$$

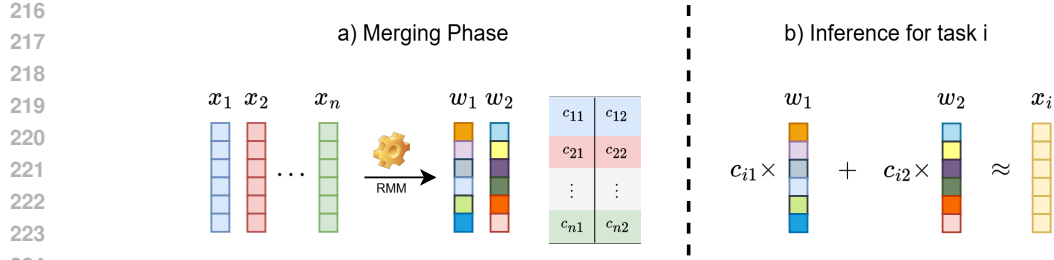


Figure 3: Overview of the proposed RMM framework. **Left:** in the merging phase, RMM computes a compact set of orthonormal basis directions (two, in this illustration) together with the task-specific coefficients that reconstruct each model from them. **Right:** at inference, target model i is reconstructed by linearly combining the shared basis vectors with its own coefficients, recovering the individual adapter on demand.

in which the two low-rank factors are merged independently and then recomposed.

Although both formulations are mathematically valid, the combined strategy is untenable in practice on storage grounds. Retaining the merged update in combined form requires an $m \times d$ matrix, whereas the separate strategy stores only low-rank factors of the size of $A_i^l \in \mathbb{R}^{m \times r}$ and $B_i^l \in \mathbb{R}^{r \times d}$. Since $r(m + d) \ll md$ whenever $r \ll \min(m, d)$, the combined approach reintroduces exactly the overhead that compression was meant to eliminate. Concretely, for a layer with $m = d = 1024$ and rank $r = 16$, the combined form stores a full update of $md = 1,048,576$ parameters while the separate form stores only $r(m + d) = 16 \times (1024 + 1024) = 32,768$ parameters—a **32 $\times$ reduction in storage**. Because the combined approach thus forfeits the entire benefit of low-rank compression, we adopt the separate strategy as the only practically viable option for merging compressed models and use it throughout.

Using this separate strategy, we applied several established merging baselines to low-rank compressed models across a range of compression ratios. We took RoBERTa-base (Zhuang et al., 2021) as the pretrained backbone and evaluated on five GLUE tasks (Wang et al., 2018); full experimental details are given in Appendix A.3. As Figures 1 and 2 make plain, merging compressed models induces a *catastrophic drop* in average task performance relative to merging their full-rank counterparts. We attribute this degradation to two mechanisms that act in concert: (i) **Amplified task interference**: compression discards much of the representational capacity, leaving low-rank models markedly more susceptible to destructive interference when their updates are superimposed; (ii) **Broken coupling between A and B** : conventional merging treats the low-rank factors A_i^l and B_i^l as independent objects, disregarding that only their product $A_i^l B_i^l$ is meaningful, and thereby introduces structural misalignment into the merged updates. Acting together, these mechanisms severely impair merged low-rank models and point to a deeper incompatibility between traditional merging and compressed representations—one that no amount of coefficient tuning within the single-model formulation can resolve.

This diagnosis yields a decisive conclusion: under conventional strategies, no single merged model can reliably recover the task-specific performance of the original low-rank adapters. It also suggests a constructive alternative: *rather than forcing all updates into one representation, can we reconstruct the original compressed models—or closely approximate them—by retaining a compact set of basis models whose cost is equivalent to only a few adapters and can be tuned to the available storage budget?* Pursuing this question leads to an alternative design space in which merging produces a *reconstruction-capable model space*. We formalize this perspective in the next section and introduce our method, *Reversible Model Merging* (RMM).

5 REVERSIBLE MODEL MERGING

The empirical failures documented above expose a fundamental limitation: no single merged model can faithfully preserve the task-specific performance of all low-rank compressed models simultaneously. RMM responds by abandoning the goal of collapsing all models into one set of weights.

In its place, RMM constructs a small set of shared components—basis models—from which every original task model can be accurately reconstructed.

Rather than compressing all n task-specific models into a single merged solution $\hat{\theta}_{\mathcal{M}}$, RMM retains a compact set of p merged components with $2 \leq p \ll n$. These components define a low-dimensional subspace that spans the essential variation across tasks, enabling each model to be recovered efficiently on demand. The formulation makes the performance–memory trade-off explicit and controllable: by tuning the single hyperparameter p , a practitioner selects any operating point between aggressive consolidation and near-lossless reconstruction.

For each layer l , we operate on the r -dimensional vectors that constitute the low-rank factors—the rows of A_i^l and the columns of B_i^l , where r is the compression rank. We designate each such row or column an independent *task vector* and its row or column index a *task vector position*. For a fixed position, we assemble the corresponding task vectors across all n models and denote the vector contributed by model i at that position as $\mathbf{x}_i \in \mathbb{R}^r$. RMM is then applied independently at each position: it compresses the set $\{\mathbf{x}_i\}_{i=1}^n$ by projecting these vectors onto a shared basis from which they can subsequently be reconstructed. Because each layer contributes $m + d$ positions in total— m rows from A_i^l and d columns from B_i^l —the same procedure is executed separately at every one of them.

Concretely, at each position we seek a set of basis vectors $\{\mathbf{w}_1, \dots, \mathbf{w}_p\}$ with each $\mathbf{w}_j \in \mathbb{R}^r$ for $1 \leq j \leq p$. Their span defines a subspace $\mathbf{W} = [\mathbf{w}_1 \dots \mathbf{w}_p] \in \mathbb{R}^{r \times p}$ onto which each task vector $\mathbf{x}_i$ is projected, subject to the requirement that the projection reconstruct $\mathbf{x}_i$ as accurately as possible. We further introduce the coefficient matrix $\mathbf{C} \in \mathbb{R}^{n \times p}$, whose i -th row $[c_{i1}, \dots, c_{ip}]$ holds the scalar weights that linearly combine the basis vectors to reconstruct $\mathbf{x}_i$. The design objective is to determine $\mathbf{W}$ and $\mathbf{C}$ that jointly minimize the aggregate reconstruction error over all n task vectors, which we formalize as

$$\min_{\mathbf{W}, \mathbf{C}} \sum_{i=1}^n \left\| \mathbf{x}_i - \sum_{j=1}^p c_{ij} \mathbf{w}_j \right\|_2^2. \quad (3)$$

This representation stores rp parameters for the basis and np for the coefficients, $p(r+n)$ in total—a substantial reduction from the rn parameters needed to retain all task vectors directly whenever $p \ll \min(r, n)$. The hyperparameter p thus directly governs the performance–storage trade-off, and Figure 3 illustrates the construction for $p = 2$. Collecting the task vectors row-wise, the objective in equation 3 can be written compactly in matrix form as

$$\min_{\mathbf{W}, \mathbf{C}} \left\| \mathbf{X} - \mathbf{C}\mathbf{W}^\top \right\|_F^2, \quad (4)$$

where $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$ stacks all task vectors as rows. For any fixed $\mathbf{W}$, the inner minimization over $\mathbf{C}$ is an ordinary least-squares problem with the closed-form solution $\mathbf{C}^* = \mathbf{X}\mathbf{W}(\mathbf{W}^\top \mathbf{W})^{-1}$. Restricting the basis to be orthonormal ($\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$ with $p < r$) collapses this to $\mathbf{C}^* = \mathbf{X}\mathbf{W}$ and reduces the joint objective to the single-variable program

$$\min_{\mathbf{W}} \left\| \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top \right\|_F^2 \quad \text{s.t.} \quad \mathbf{W}^\top \mathbf{W} = \mathbf{I}_p. \quad (5)$$

Under this formulation we store the basis $\mathbf{W} \in \mathbb{R}^{r \times p}$ together with $\mathbf{X}\mathbf{W} \in \mathbb{R}^{n \times p}$, the latter being exactly the coefficient matrix $\mathbf{C}$ of equation 4.

Theorem 1 (Optimal Solution to equation 5). *Let $\mathbf{X} \in \mathbb{R}^{n \times r}$ be zero-centered and $p < \min\{r, n\}$. The optimal basis $\mathbf{W}^*$ that minimizes equation 5 is given by the top- p eigenvectors of the sample covariance matrix $\mathbf{X}^\top \mathbf{X}$. Equivalently, $\mathbf{W}^*$ consists of the top- p right singular vectors of $\mathbf{X}$.*

The proof of Theorem 1 is provided in Appendix A.1. This closed-form characterization is the crux of RMM: it renders basis selection data-free and optimal in the reconstruction sense, obviating the iterative optimization and data access that competing methods require. We now describe the two phases of RMM that operationalize it.

5.1 MERGING PHASE

The merging phase constructs the compact shared-basis representation of all task-specific models and is performed offline, prior to any inference. For each layer and each task vector position (a fixed

row of A_i^l or column of B_i^l), we assemble the task vectors of all n models into $X = [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$. We compute the mean vector $\mu = \frac{1}{n} \sum_{i=1}^n x_i$ and zero-center the data ($X \leftarrow [x_1 - \mu \dots x_n - \mu]^\top$), satisfying the precondition of Theorem 1. We then factorize the centered matrix by SVD, $X = U \Sigma V^\top$, and take its top- p right singular vectors as the optimal orthonormal basis for equation 5, $W^* = V_{[:,1:p]} \in \mathbb{R}^{r \times p}$. Finally, we form the coefficient matrix $C^* = XW^* \in \mathbb{R}^{n \times p}$, whose i -th row $[c_{i1}, \dots, c_{ip}]$ supplies exactly the weights that reconstruct x_i from the shared basis.

On completing this phase we discard the original task vectors $\{x_i\}_{i=1}^n$ and retain, per position, only the basis W^* , the coefficients C^* , and the mean vector μ . This reduces the per-position footprint from the rn parameters needed to store all task vectors to $p(r+n)+r$, the additive r accounting for the retained mean. Because the phase runs entirely offline and without task data, its cost is incurred once and does not affect inference-time efficiency.

5.2 INFERENCE PHASE

Consistent with common practice in model merging, we assume the target task for each input is either specified by the user or resolved by an oracle router that maps the input to the most suitable model index. Given the target index i , the inference phase reconstructs its task vector x_i as $\hat{x}_i$ from the compact per-position representation (W^*, C^*, μ) produced during merging. Reassembling the reconstructed vectors $\hat{x}_i$ across all positions—the rows for A_i^l and the columns for B_i^l —recovers the full low-rank factors of model i .

Recall that, at each position, $W^* \in \mathbb{R}^{r \times p}$ holds the orthonormal basis, $C^* \in \mathbb{R}^{n \times p}$ the task-specific coefficients, and $\mu \in \mathbb{R}^r$ the mean across task vectors. Reconstruction then amounts to a linear combination of the basis directions with model i ’s coefficients, re-centered by the mean μ :

$$\hat{x}_i = \sum_{j=1}^p c_{ij}^* w_j^* + \mu = C_{[i,:]}^* W^{*\top} + \mu.$$

The resulting $\hat{x}_i \in \mathbb{R}^r$ approximates the original task vector x_i discarded during merging. Carried out independently at every position and layer, this reconstruction yields the complete low-rank delta $\hat{\Delta}_i$ of task-specific model i . Algorithm 1 states the full merging and inference procedures in detail. We next evaluate RMM empirically and benchmark it against standard baselines.

6 EXPERIMENTS

To assess RMM comprehensively, we conduct extensive experiments spanning multiple compression settings, task suites, and model architectures. Our evaluation is designed to answer two questions: whether RMM preserves the performance of low-rank task-specific models while reducing overall storage, and how it compares to conventional merging baselines across these regimes.

6.1 EXPERIMENTAL SETUP

We first evaluate RMM on eight RoBERTa-base models fine-tuned for eight distinct tasks (Zhuang et al., 2021). Each model is compressed independently with either PT-SVD or LoRA at target ranks $r \in \{16, 32, 64, 128\}$, after which RMM is applied to the resulting low-rank models and compared against three standard merging baselines: Task Arithmetic (TA) (Ilharco et al., 2023), TIES-merging (Yadav et al., 2023), and DARE (Yu et al., 2024). As an upper bound we additionally report the performance of the individual compressed models before merging. To probe generality across backbones, modalities, and task counts, Appendix A.2 reports further experiments on OPT-1.3b (Zhang et al., 2022) and on ViT-B/32 (Radford et al., 2021) for vision tasks.

We follow the GLUE benchmark protocol (Wang et al., 2018) and evaluate each task with its standard metric: QNLI (question–answering inference), MRPC (paraphrase detection), SST-2 (sentiment classification), MNLI (multi-genre natural language inference), QQP (duplicate question detection), and RTE (textual entailment) by accuracy; CoLA (linguistic acceptability) by Matthews correlation; and STS-B (semantic textual similarity) by Pearson correlation. The average across tasks serves as the overall performance measure. Alongside accuracy, we quantify the relative storage cost of each method as the fraction of parameters it retains relative to storing all individual

Algorithm 1 Reversible Model Merging (RMM)

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1: Merging Phase:
2: Require: Low-rank deltas  $A_i^l$  and  $B_i^l$ , compression rank  $r$ , number of basis vectors  $p$ 
3: for  $l = 1$  to  $L$  do
4:   for each task vector position (each row in  $A_i^l$  or each column in  $B_i^l$ ) do
5:     Gather task vectors  $x_i$  of each model  $i$  ( $1 \leq i \leq n$ )
6:      $X \leftarrow [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ 
7:      $\mu \leftarrow \sum_{i=1}^n x_i \in \mathbb{R}^r$ 
8:      $X \leftarrow [x_1 - \mu \dots x_n - \mu]^\top$ 
9:      $U, \Sigma, V^\top \leftarrow \text{SVD}(X)$ 
10:     $W^* \leftarrow V_{[:,1:p]} \in \mathbb{R}^{r \times p}$ 
11:     $C^* \leftarrow XW^* \in \mathbb{R}^{n \times p}$ 
12:    Store  $(W^*, C^*, \mu)$  and discard task vectors  $\{x_1 \dots x_n\}$ 
13:   end for
14: end for
15: Inference Phase:
16: Require: target task  $i$ , stored representation  $(W^*, C^*, \mu)$  for each task vector position
17: for  $l = 1$  to  $L$  do
18:   for each task vector position do
19:      $\hat{x}_i = C_{[i,:]}^* W^{*\top} + \mu$ 
20:   end for
21:    $\hat{A}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all row positions})$ 
22:    $\hat{B}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all column positions})$ 
23: end for
24:  $\hat{\Delta}_i = \{\hat{A}_i^l \hat{B}_i^l\}_{l=1}^L$ 
25: Inference with the reconstructed model  $\theta_{pre} + \hat{\Delta}_i$ 

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task-specific adapters. Baselines that store a single merged model incur a fixed cost of $\frac{r}{rn} = \frac{1}{n}$, whereas RMM with p basis components incurs $\frac{p(r+n)+r}{rn}$, as derived in Section 5.1; reporting both axes makes the accuracy–storage trade-off directly legible in the tables that follow.

6.2 RESULTS AND DISCUSSION

Tables 1 and 2 report RMM against the baselines under PT-SVD and LoRA compression, respectively. Across both compression schemes and all four target ranks $r \in \{16, 32, 64, 128\}$, RMM outperforms TA, TIES, and DARE by a wide and uniform margin. At the lower ranks ($r = 16, 32$) the single-model baselines fail catastrophically, attaining only 31–42% average score and collapsing almost entirely on the structurally sensitive tasks CoLA and STS-B. RMM with $p = 2$ already recovers a substantial share of the lost performance (e.g., 57.24% under PT-SVD and 51.41% under LoRA at $r = 16$), and RMM with $p = 3$ narrows the gap to the individual compressed models further still (up to 72.22% and 58.64% at $r = 16$). At the higher ranks ($r = 64, 128$), where compression error is smaller, RMM continues to dominate: with $p = 3$ it reaches 64.34% and 60.58% under PT-SVD and 55.40% and 61.35% under LoRA, while the single-model baselines remain stalled near 40%. Even the more economical $p = 2$ configuration retains a clear advantage throughout, confirming that the reversible formulation structurally averts the destructive interference that cripples prior methods rather than merely mitigating it.

We further observe that the relative gain of RMM over the baselines narrows as the rank grows. A natural explanation is that higher-rank task representations occupy higher-dimensional, more expressive, and less mutually aligned subspaces, which are inherently harder to compress into a shared low-dimensional basis; the effect is expected and does not reverse the ranking, as RMM remains ahead at every rank. Taken together, the results establish a consistent advantage for RMM across both compression regimes and all ranks: it preserves high task performance in exchange for additional storage, and the number of reversible components p exposes this accuracy–storage trade-off as an explicit, tunable dial. Crucially, the added cost is confined to **offline memory**—typically inexpen-

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	77.98	58.05	90.59	83.80
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.19	31.66
	TIES	13%	50.56	34.78	49.08	31.82	36.82	47.29	0.00	3.75	31.76
	DARE	13%	50.80	33.51	49.08	31.82	36.82	47.29	0.00	5.07	31.80
	RMM ($p = 2$)	50%	51.20	78.20	88.30	66.74	86.02	47.29	6.56	33.59	57.24
	RMM ($p = 3$)	69%	88.83	86.26	92.89	79.58	87.18	48.74	33.85	60.46	72.22
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	78.34	59.33	90.68	84.46
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.21	31.66
	TIES	13%	50.56	35.19	49.08	31.82	36.82	47.29	0.00	3.87	31.83
	DARE	13%	50.72	33.51	49.08	31.82	36.82	47.29	0.00	3.48	31.59
	RMM ($p = 2$)	44%	50.54	74.32	55.05	62.68	87.07	47.29	0.00	23.18	50.02
	RMM ($p = 3$)	59%	84.66	86.32	92.89	79.70	88.05	47.29	23.57	44.54	68.38
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	78.70	61.11	90.72	85.10
	TA	13%	50.61	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.13	49.08	31.82	36.82	47.29	0.00	3.80	31.82
	DARE	13%	50.58	33.51	49.08	31.82	36.82	47.29	0.00	5.22	31.79
	RMM ($p = 2$)	41%	50.53	73.10	49.08	58.48	86.98	47.29	0.00	17.50	47.87
	RMM ($p = 3$)	55%	74.14	86.20	92.43	78.83	89.00	47.29	13.16	33.70	64.34
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	79.78	60.16	90.74	85.30
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.54	49.08	31.82	36.82	47.29	0.00	3.93	31.88
	DARE	13%	50.61	34.78	49.08	31.82	36.82	47.29	0.00	2.41	31.60
	RMM ($p = 2$)	39%	50.54	70.78	49.08	54.69	84.68	47.29	0.00	14.53	46.45
	RMM ($p = 3$)	52%	59.20	85.80	92.32	77.50	89.02	47.29	6.56	26.94	60.58

Table 1: GLUE performance for merging eight RoBERTa-base models compressed with PT-SVD at ranks $r \in \{16, 32, 64, 128\}$. *No merging* is the individual-model upper bound; TA, TIES, and DARE store a single merged model, while RMM stores p reversible components. At every rank RMM ($p=3$) improves the average score over the strongest baseline by a wide margin—from roughly 40 points at $r=16$ (72.22 vs. 31.80) to roughly 29 points at $r=128$ (60.58 vs. 31.88)—and RMM ($p=2$) already surpasses all baselines, at the higher storage fractions reported in the Storage column.

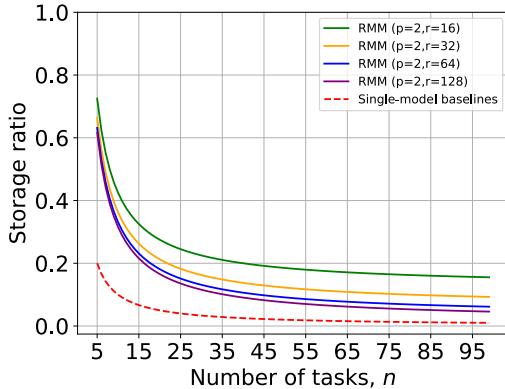


Figure 4: Scalability of RMM ($p = 2$): storage ratio—the memory required as a percentage of storing all models—as a function of the number of tasks. The ratio falls steadily as tasks accumulate, evidencing sublinear storage growth.

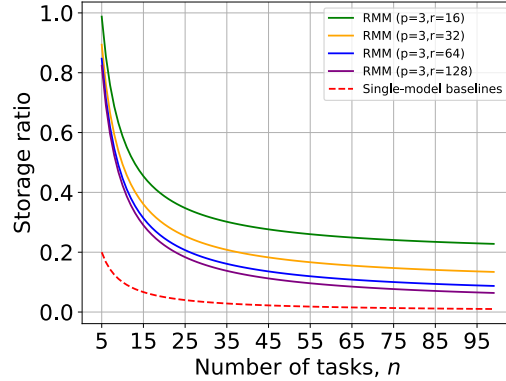


Figure 5: Scalability of RMM ($p = 3$): storage ratio—the memory required as a percentage of storing all models—as a function of the number of tasks. As with $p = 2$, the ratio declines with the task count, so the higher reconstruction fidelity of $p = 3$ retains the same favorable scaling.

sive, CPU-resident storage—whereas task performance governs runtime quality. In such settings, a modest increase in CPU-side storage is a favorable exchange for markedly higher accuracy.

6.3 SCALABILITY OF RMM

We now examine how RMM scales as the number of models n grows. Figures 4 and 5 plot the storage ratio, relative to retaining all individual task-specific models, against the number of tasks

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	68.23	53.92	88.71	77.71
	TA	13%	50.59	33.51	72.13	34.79	64.46	47.29	0.00	21.81	40.57
	TIES	13%	50.63	33.68	74.31	35.64	67.45	47.29	0.00	26.38	41.92
	DARE	13%	50.63	33.51	73.74	35.97	65.35	47.29	0.00	21.62	41.01
	RMM ($p = 2$)	50%	51.13	34.14	90.71	60.48	84.41	47.29	0.00	43.14	51.41
	RMM ($p = 3$)	69%	55.92	38.14	93.12	82.95	87.30	47.29	6.56	57.81	58.64
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	62.82	53.12	88.53	76.54
	TA	13%	50.91	68.06	54.70	33.77	65.50	55.96	0.00	-0.48	41.05
	TIES	13%	51.03	68.06	56.77	34.24	66.27	58.12	0.00	1.02	41.94
	DARE	13%	50.96	64.64	63.07	35.12	63.66	55.60	0.00	1.01	41.76
	RMM ($p = 2$)	44%	52.08	65.39	81.54	62.46	77.06	57.40	0.00	9.95	50.73
	RMM ($p = 3$)	59%	54.62	69.97	92.20	81.29	85.97	57.40	4.64	24.08	58.77
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	67.15	58.31	88.70	78.25
	TA	13%	54.75	68.52	77.29	37.24	36.82	47.29	0.00	-3.53	39.80
	TIES	13%	54.82	68.75	80.73	37.67	36.82	47.29	0.00	-2.48	40.45
	DARE	13%	53.71	68.58	78.56	38.33	36.82	47.29	0.00	-4.28	39.88
	RMM ($p = 2$)	41%	55.74	69.28	87.27	42.81	77.72	47.29	0.00	4.91	48.13
	RMM ($p = 3$)	55%	59.07	70.14	90.83	79.11	82.06	47.29	0.00	14.70	55.40
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	57.04	59.35	89.68	77.71
	TA	13%	51.49	69.22	54.47	37.16	63.27	47.29	0.00	-1.74	40.15
	TIES	13%	51.91	68.93	57.34	37.81	63.36	47.29	0.00	-0.25	40.80
	DARE	13%	51.97	68.99	58.60	35.64	63.33	47.29	0.00	-0.44	40.67
	RMM ($p = 2$)	39%	57.18	67.83	88.53	49.54	84.64	47.29	0.00	28.00	52.88
	RMM ($p = 3$)	52%	59.13	39.19	92.09	80.08	86.39	47.29	15.45	71.18	61.35

Table 2: GLUE performance for merging eight RoBERTa-base models compressed with LoRA at ranks $r \in \{16, 32, 64, 128\}$. Column conventions match Table 1. RMM ($p=3$) again leads at every rank, exceeding the strongest baseline average by roughly 15–21 points (e.g., 58.64 vs. 41.92 at $r=16$ and 61.35 vs. 40.80 at $r=128$), and RMM ($p=2$) remains ahead of all single-model baselines throughout.

for $p = 2$ and $p = 3$, respectively. Both curves confirm favorable scaling: the relative storage cost decreases markedly as tasks accumulate. This behavior is the central operational benefit of reversible merging—storage grows sublinearly in the number of task-specific models—and it follows directly from the design. Once the shared basis is fixed, incorporating a new task adds only a single row to the coefficient matrix, whereas conventional approaches must either commit to one merged model, sacrificing performance, or store every task-specific model independently, incurring strictly linear growth. Consequently, RMM is particularly well suited to large-scale multi-task, continual, and federated settings that require managing hundreds of models efficiently without compromising per-task performance.

7 CONCLUSION

We introduced a principled framework for reversible model merging that redefines how multiple fine-tuned models are stored, merged, and reconstructed. Rather than forcing all task-specific knowledge into a single merged model, RMM relaxes this constraint by retaining a compact set of shared components—substantially fewer than the number of original models, yet strictly more expressive than a single aggregate—from which each model can be recovered by linear combination. Grounded in a closed-form, data-free basis-selection solution, this design reconstructs task-specific models accurately while sharply reducing storage overhead, and it exposes a controllable accuracy–storage trade-off that is especially valuable for low-rank compressed models. By representing fine-tuned models through compact bases and reversible mappings, we demonstrate that models can be merged, reconstructed, and deployed at high accuracy under strict memory budgets, with the degree of compression and reconstruction fidelity tuned to the demands of the application.

More broadly, by formalizing merging as a structured, reversible process, our framework moves decisively beyond heuristic, one-shot merging. This perspective not only enlarges the design space of efficient multi-task deployment but also charts promising directions for future research on performance-aware compression and reconstruction.

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A APPENDIX

A.1 PROOF OF THEOREM 1

Let $\mathbf{A} = \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top$ denote the reconstruction residual. By the definition of the Frobenius norm, $\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}^\top \mathbf{A})$, where $\text{Tr}(\cdot)$ is the trace of a square matrix. Expanding the residual yields

$$\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 = \text{Tr}((\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)).$$

Expanding the argument of the trace gives

$$\begin{aligned} (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) &= (\mathbf{X}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top)(\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) \\ &= \mathbf{X}^\top \mathbf{X} - \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X} \\ &\quad + \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top. \end{aligned}$$

Taking the trace of this expansion and invoking the cyclic property of the trace together with the orthonormality constraint $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$, we obtain

$$\begin{aligned} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}). \end{aligned}$$

Because $\text{Tr}(\mathbf{X}^\top \mathbf{X})$ does not depend on $\mathbf{W}$, minimizing $\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2$ is equivalent to maximizing $\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W})$. Consequently,

$$\min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \iff \max_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}).$$

By the Rayleigh quotient theorem (Horn & Johnson, 2012), this trace is maximized by taking the columns of $\mathbf{W}$ to be the top- p eigenvectors of $\mathbf{X}^\top \mathbf{X}$. This is exactly the principal component analysis (PCA) solution: it identifies the p -dimensional subspace that maximizes the projected variance and, equivalently, minimizes the squared reconstruction error. Moreover, writing the SVD of $\mathbf{X}$

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	BOOLQ	Average
16	No merging	100%	91.74	71.54	95.30	84.19	88.14	71.47	83.73
	Task-Arithmetic	17%	50.34	37.16	58.49	34.43	63.30	41.87	47.60
	TIES	17%	49.79	36.75	61.24	35.57	63.55	40.67	47.93
	DARE	17%	49.95	37.68	56.08	34.51	63.39	42.78	47.40
	RMM ($p = 2$)	63%	88.12	44.00	93.58	73.06	85.63	59.69	74.01
	RMM ($p = 3$)	85%	91.58	49.04	95.64	80.31	87.52	65.50	78.27
32	No merging	100%	92.06	72.17	95.41	84.60	89.17	72.45	84.31
	Task-Arithmetic	17%	50.28	37.22	58.94	34.57	63.33	41.93	47.71
	TIES	17%	49.77	36.93	61.93	35.69	63.65	40.73	48.12
	DARE	17%	50.39	40.87	56.19	34.96	63.59	43.33	48.22
	RMM ($p = 2$)	56%	86.66	41.91	91.74	71.75	85.68	56.39	72.36
	RMM ($p = 3$)	76%	91.85	44.81	94.84	80.21	88.43	63.52	77.28
64	No merging	100%	92.26	72.12	95.41	84.64	90.07	72.84	84.56
	Task-Arithmetic	17%	50.28	37.28	59.06	34.56	63.34	42.08	47.77
	TIES	17%	49.75	37.04	61.81	35.84	63.74	40.76	48.16
	DARE	17%	50.16	40.52	58.83	35.59	63.35	42.54	48.50
	RMM ($p = 2$)	53%	83.87	42.72	89.45	69.76	85.70	53.43	70.82
	RMM ($p = 3$)	71%	92.07	41.97	94.50	79.70	88.73	61.96	76.49
128	No merging	100%	92.35	72.41	95.41	84.67	90.93	73.09	84.81
	Task-Arithmetic	17%	50.32	37.39	58.94	34.63	63.37	42.08	47.79
	TIES	17%	49.79	37.04	62.04	35.99	63.74	40.55	48.19
	DARE	17%	50.03	38.14	57.91	34.72	63.31	42.11	47.70
	RMM ($p = 2$)	52%	81.38	42.78	84.75	67.66	83.22	51.99	68.63
	RMM ($p = 3$)	69%	92.28	41.80	93.69	79.42	88.76	59.69	75.94

Table 3: GLUE performance for merging six OPT-1.3b models compressed with PT-SVD at ranks $r \in \{16, 32, 64, 128\}$. Both RMM variants surpass every single-model baseline at all ranks; with $p=3$, RMM recovers average scores that closely track the individual-model upper bound (e.g., 78.27 vs. 83.73 at $r=16$) while retaining only a fraction of the parameters needed to store all adapters. This confirms that the gains observed on RoBERTa transfer to a substantially larger decoder-only backbone.

as $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ gives $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top$, so the eigenvectors of $\mathbf{X}^\top \mathbf{X}$ coincide with the right singular vectors of $\mathbf{X}$ (the columns of $\mathbf{V}$). The optimal basis can therefore be read off directly from the top- p right singular vectors of $\mathbf{X}$ —an SVD-based route that is numerically more stable, and frequently computationally preferable, to explicitly forming $\mathbf{X}^\top \mathbf{X}$, particularly when r is large. This is precisely the construction used in the merging phase of Algorithm 1.

A.2 ADDITIONAL RESULTS

To probe the generality of RMM beyond RoBERTa, we present extended results for two additional pretrained backbones spanning different architectures and modalities: OPT-1.3b and ViT-B/32. For **OPT-1.3b**, we evaluate on six natural language understanding tasks drawn from GLUE and related benchmarks—QNLI, MRPC, SST-2, MNLI, QQP, and BoolQ (Clark et al., 2019). For **ViT-B/32**, we turn to vision classification across seven diverse datasets: MNIST (LeCun et al., 1998), GTSRB (Stallkamp et al., 2011), SVHN (Netzer et al., 2011), Cars (Krause et al., 2013), EuroSAT (Helber et al., 2019), DTD (Cimpoi et al., 2014), and SUN397 (Xiao et al., 2010).

Table 3 reports the six OPT-1.3b models compressed with PT-SVD. Both RMM variants consistently outperform the baselines, and RMM at $p = 3$ narrows the gap to the original individual models while requiring substantially fewer parameters, confirming that the advantages observed on RoBERTa carry over to a much larger decoder-only language model. Table 4 reports the seven ViT-B/32 models, where the benefit of RMM is more pronounced still: at $p = 3$ it attains an average accuracy less than 3% below that of the original models while reducing storage to only 59% of the cost of retaining all models at $r = 128$. Together, these results show that RMM faithfully reconstructs the original compressed models across architectures and modalities while delivering significant storage efficiency, evidencing that its effectiveness is not specific to any single backbone or task family.

A.3 RESULTS FOR 5 TASKS

The motivating study of Section 4 used RoBERTa-base as the pretrained backbone and evaluated the merging baselines on five GLUE tasks (QNLI, MRPC, SST-2, MNLI, QQP); Figures 1 and 2

r	Method	Storage	MNIST	GTSRB	SVHN	Cars	EuroSAT	DTD	SUN397	Average
16	No merging	100%	99.61	97.22	97.06	69.20	99.67	75.21	72.72	87.24
	Task-Arithmetic	14%	47.19	28.29	38.64	56.27	57.52	41.38	62.92	47.46
	TIES	14%	60.14	34.52	49.03	58.55	61.70	43.67	62.87	52.93
	DARE	14%	53.80	31.00	44.96	57.60	56.93	43.99	62.42	50.10
	RMM ($p = 2$)	55%	99.16	88.52	94.89	66.61	97.44	60.64	67.86	82.16
	RMM ($p = 3$)	76%	99.58	95.65	96.54	68.30	98.93	67.18	69.50	85.10
32	No merging	100%	99.65	98.45	97.35	73.04	99.78	78.72	75.52	88.93
	Task-Arithmetic	14%	47.90	28.31	38.68	56.24	57.52	41.38	63.00	47.58
	TIES	14%	61.25	35.28	49.95	58.80	62.41	44.04	63.05	53.54
	DARE	14%	53.05	31.04	46.34	58.54	57.74	42.87	62.87	50.35
	RMM ($p = 2$)	49%	98.98	88.23	94.29	68.32	96.81	61.91	68.36	82.41
	RMM ($p = 3$)	67%	99.50	96.53	96.68	70.90	98.93	70.43	71.23	86.31
64	No merging	100%	99.67	98.68	97.41	75.76	99.85	79.15	77.73	89.75
	Task-Arithmetic	14%	47.99	28.30	38.71	56.21	57.59	41.38	63.22	47.63
	TIES	14%	61.83	35.93	50.28	59.08	63.07	44.15	63.17	53.93
	DARE	14%	54.78	32.79	45.05	58.23	61.30	42.87	62.77	51.11
	RMM ($p = 2$)	46%	98.46	84.43	92.83	69.36	95.59	62.39	69.77	81.83
	RMM ($p = 3$)	62%	99.45	96.14	96.33	72.84	98.56	72.07	73.85	87.03
128	No merging	100%	99.68	98.75	97.46	76.99	99.85	79.04	78.59	90.05
	Task-Arithmetic	14%	48.00	28.31	38.73	56.21	57.63	41.44	63.35	47.67
	TIES	14%	62.08	36.36	50.48	59.23	63.41	44.68	63.25	54.21
	DARE	14%	55.16	32.07	45.47	58.55	61.85	43.09	62.85	51.29
	RMM ($p = 2$)	44%	97.22	78.56	90.63	70.22	94.00	61.49	71.03	80.45
	RMM ($p = 3$)	59%	99.28	94.59	95.95	74.24	98.19	71.86	76.02	87.16

Table 4: Vision performance for merging seven ViT-B/32 models compressed with PT-SVD at ranks $r \in \{16, 32, 64, 128\}$. The reversible formulation is especially effective in the vision setting: RMM ($p=3$) tracks the individual-model upper bound closely across all ranks (e.g., 87.16 vs. 90.05 at $r=128$), remaining within roughly three points while storing only 59% of the parameters, whereas the single-model baselines lose tens of points. This demonstrates that RMM extends beyond language to a distinct architecture and modality.

summarize those results at a glance. For completeness, we provide the underlying per-task numbers here: Table 5 accompanies Figure 1 and reports the LoRA fine-tuned models together with the improvements delivered by RMM, while Table 6 accompanies Figure 2 and reports the PT-SVD compressed models. These tables make explicit that, even in this five-task regime, RMM recovers near-individual-model performance where every baseline collapses.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	82.16
	Task-Arithmetic	20%	50.61	33.51	73.85	34.23	64.58	51.36
	TIES	20%	50.65	33.51	75.57	35.83	66.97	52.51
	DARE	20%	50.61	33.51	79.36	34.70	63.40	52.32
	RMM ($p = 2$)	73%	59.64	68.64	93.00	82.68	86.89	78.17
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	81.57
	Task-Arithmetic	20%	51.18	67.30	58.83	34.33	68.56	56.04
	TIES	20%	51.36	68.17	59.52	34.76	71.46	57.05
	DARE	20%	50.83	64.87	68.12	35.54	68.93	57.66
	RMM ($p = 2$)	66%	57.72	67.54	92.55	81.80	86.13	77.15
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	82.37
	Task-Arithmetic	20%	54.74	68.93	80.28	35.70	36.82	55.29
	TIES	20%	55.26	68.93	78.78	36.76	36.82	55.31
	DARE	20%	54.71	68.81	74.54	38.72	36.82	54.72
	RMM ($p = 2$)	63%	59.82	67.48	91.51	83.21	84.41	77.29
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	83.12
	Task-Arithmetic	20%	52.52	68.70	54.82	37.83	63.46	55.47
	TIES	20%	52.55	69.04	57.80	38.91	63.72	56.40
	DARE	20%	55.04	69.62	54.47	38.92	64.12	56.43
	RMM ($p = 2$)	62%	60.21	34.96	91.86	78.59	86.65	70.45

Table 5: GLUE performance for merging five RoBERTa-base models compressed with LoRA at ranks $r \in \{16, 32, 64, 128\}$ (per-task detail for Figure 1). RMM ($p=2$) lifts the average score far above every baseline—e.g., 78.17 vs. at most 52.51 at $r=16$ —approaching the individual-model upper bound while the baselines never exceed the upper 50s.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	88.76
	Task-Arithmetic	20%	53.03	35.19	49.08	31.82	36.82	41.19
	TIES	20%	55.46	54.32	49.08	31.82	37.21	45.58
	DARE	20%	53.69	58.67	49.08	31.82	37.08	46.07
	RMM ($p = 2$)	73%	79.72	85.16	90.83	76.58	86.09	83.68
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	89.46
	Task-Arithmetic	20%	53.23	35.59	49.08	31.82	36.82	41.31
	TIES	20%	55.78	55.19	49.08	31.82	37.62	45.90
	DARE	20%	55.89	33.74	49.20	31.82	36.85	41.50
	RMM ($p = 2$)	66%	63.87	82.78	91.28	77.12	87.05	80.42
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	90.05
	Task-Arithmetic	20%	53.49	35.88	49.08	31.82	36.82	41.42
	TIES	20%	56.14	55.94	49.08	31.82	38.08	46.21
	DARE	20%	53.51	33.68	49.08	31.82	36.82	40.98
	RMM ($p = 2$)	63%	56.12	81.22	91.06	76.74	88.07	78.64
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	90.34
	Task-Arithmetic	20%	53.60	36.06	49.08	31.82	36.82	41.48
	TIES	20%	56.40	56.35	49.08	31.82	38.61	46.45
	DARE	20%	50.52	34.96	49.08	31.82	36.92	40.66
	RMM ($p = 2$)	62%	54.11	79.19	90.60	74.49	87.46	77.17

Table 6: GLUE performance for merging five RoBERTa-base models compressed with PT-SVD at ranks $r \in \{16, 32, 64, 128\}$ (per-task detail for Figure 2). The pattern mirrors the LoRA case: RMM ($p=2$) reaches 83.68 at $r=16$ against a best baseline of 46.07, and stays close to the individual-model upper bound across all ranks while the baselines languish in the low-to-mid 40s.